

A Portable and a Scalable Multi-Channel Wireless Recording System for Wearable Electromyometrial Imaging

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Abstract—Electromyometrial imaging (EMMI) technology has emerged as one of the promising technology that can be used for non-invasive pregnancy risk stratification and for preventing complications due to pre-term birth. Current EMMI systems are bulky and require a tethered connection to desktop instrumentation, as a result, the system cannot be used in non-clinical and ambulatory settings. In this article, we propose an approach for designing a scalable, portable wireless EMMI recording system that can be used for in-home and remote monitoring. The wearable system uses a non-equilibrium differential electrode multiplexing approach to enhance signal acquisition bandwidth and to reduce the artifacts due to electrode drifts, amplifier 1/f noise, and bio-potential amplifier saturation. A combination of active shielding, a passive filter network, and a high-end instrumentation amplifier ensures sufficient input dynamic range (> 100 dB) such that the system can simultaneously acquire different bio-potential signals like maternal electrocardiogram (ECG) in addition to the EMMI electromyogram (EMG) signals. We show that the switching artifacts and the channel cross-talk introduced due to non-equilibrium sampling can be reduced using a compensation technique. This enables the system to be potentially scaled to a large number of channels without significantly increasing the system power dissipation. We demonstrate the feasibility of the proposed approach in a clinical setting using an 8-channel battery-powered prototype

which dissipates less than $8 \mu\text{W}$ per channel for a signal bandwidth of 1 KHz.

Index Terms—Electromyometrial imaging, multi-channel instrumentation, ECG, wearable, pre-term labor.

I. INTRODUCTION

PRETERM birth is a significant cost burden in perinatology, and to date robust prediction of preterm labor and birth has been a major challenge [1], [2], [3]. This in part is due to our limited ability to continuously monitor the initiation and propagation of uterine contractions at sufficient spatial and temporal resolution. Recently, electromyometrial imaging (EMMI) has emerged as a promising technology to monitor the progression of uterine contractions in real-time by measuring electromyogram (EMG) signals using body-surface electrodes placed at different spatial locations on the patient's abdomen [4], [5], [6], [7]. The spatial and temporal EMG information is then used to generate body-surface electrical maps which can then be used for diagnostics and prognostics [8], [9]. Unfortunately, current EMMI systems are tethered, relatively bulky, and hence can only be used in clinical settings; whereas the greatest impact of the EMMI technology is in low-resource and portable settings, like in-home or at a remote site. The concept is illustrated in Fig. 1 which shows a portable EMMI device (D) interfacing with an array of a flexible and wearable electrode array (C) placed over the subject's abdomen (A) and surrounding the uterine region (B). A portable and wearable device measures multi-channel biopotential signals comprising a mixture of EMGs due to uterine contractions and maternal and fetal electrocardiograms (ECGs). The acquired signal is either stored on the device or wirelessly transmitted (E) to a remote data processor which then generates the body-surface maps for visualization or for clinical interpretation. Similar to other multi-channel biopotential recording systems like electroencephalogram (EEGs) systems [10], [11], portable and wearable recording of EMMI signals in real-time presents significant engineering challenges. The magnitude of EMMI biopotentials is less than $100 \mu\text{V}$ and lies within the 0.1 Hz–10 Hz frequency range [12], [13]. This signal band also overlaps with the bandwidth of the maternal ECG signals and the skeletal EMG signals whose magnitude could be more than 10 times the EMMI signals. Furthermore,

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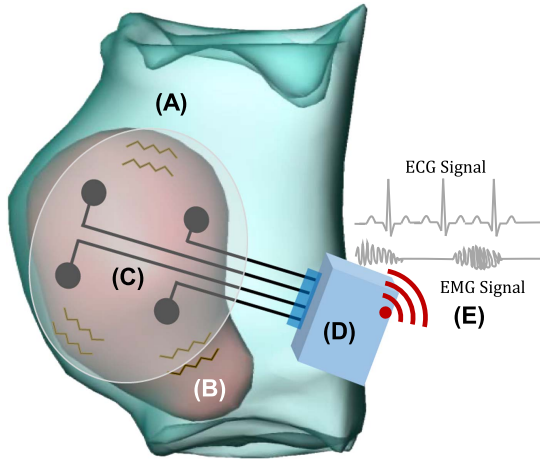


Fig. 1. Concept of a portable EMMI system comprising of an array of flexible and wearable body-surface electrodes that interface with battery-powered wireless instrumentation. The multi-channel system continuously monitors electrical signals due to uterine contraction EMG and maternal and fetal ECGs.

signals generated by motion artifacts and slow electrode drifts can also fall within the EMMI frequency band. The magnitude of the motion artifact signals could be 20-100 times larger than the EMMI signals. Thus, the amplification and the data acquisition circuitry should have a sufficient dynamic range to avoid saturation and to faithfully capture both the EMMI and ECG signals for any signal post-processing successful. Note that in this setting, the maternal ECG also serves the diagnostic purpose to verify the quality of the electrical contact between the electrode and the skin. Reducing the effect of large DC drifts (1/f noise) at the electrode interface (due to charge build-up) requires a band-pass filtering stage with low-frequency cut-off (less than 0.1 Hz).

In literature, two types of multi-channel biopotential recording systems have been proposed and are illustrated in Fig. 2. The first type is shown in Fig. 2(a) and uses independent differential amplifiers $A_1 - A_N$ to amplify the electrode signal. Each amplifier in this configuration also includes a bandpass filtering section B and an input common-mode circuit to ensure that each of the amplifiers does not saturate in the presence of electrode drifts. The output of the amplifier is then multiplexed by switches $S_1 - S_N$ to the input of a data acquisition and data processing stage. This configuration is ideal for integrated circuits (IC) implementation [14], [15], [16], [17] where a large amplifier array can be integrated within a single chipset. However, the low-frequency cut-off requirements on the bandpass filter, the low input-referred noise requirements, and the fast output settling time requirements imply that each amplifier has to be biased with a minimum tail current and hence has to dissipate a minimum amount of power. As the number of channels increases, so will the power requirements proportional to the number of amplifiers. Note that power-gating of the amplifier could be used, however, the method is limited by power-up setting-time requirements.

The second configuration shown in Fig. 2(b) alleviates some of the disadvantages by using a single, high-dynamic range and high-bandwidth instrumentation amplifier A . The differential

inputs of the electrodes are directly multiplexed at the input of the amplifier. Because each electrode has a different interface potential, switching across the electrodes emulates a chopper-like effect [18], [19], [20], [21] that alleviates DC artifacts. Furthermore, for low-bandwidth signals (EMMI and ECG), the power dissipation of this configuration is determined by the energy efficiency of the instrumentation amplifier and hence the approach can potentially be scaled to a large number of input channels. Unfortunately, this configuration is susceptible to switching artifacts (charge injection and clock feed-through) which can degrade the signal-to-noise ratio of the acquired biopotentials. Therefore, in literature, this configuration has only been used with sufficient settling time between each switching cycle which allows the electrode interface to reach electrical equilibrium. However, this reduces the effective speed of acquisition and hence, for a target bandwidth requirement limits the scalability of the approach to a large number of channels.

In this article, we investigate a non-equilibrium sampling technique based on the configuration shown in Fig. 2(b) for acquiring EMMI and ECG signals in real-time. The salient features of the proposed multi-channel data-acquisition system are:

- The approach uses a non-equilibrium electrode multiplexing technique that eliminates the latency due to settling times between channel switching.
- The approach exploits the large electrode capacitance to reduce the switching artifacts, to achieve DC blocking, and for input common-mode referencing. Note that the surface area of the recording electrodes is more than 1 cm^2 which results in electrode capacitances in the range of 100 nF-1 μF [22], [23], [24], [25].
- An active shielding technique has been used to reduce the effect of parasitic capacitances at the input and to reduce the effect of external interfering signals.
- The artifacts introduced by non-equilibrium sampling are alleviated using a frequency-domain cross-channel compensation technique.
- Using fabricated and assembled prototypes we demonstrate the functionality of an 8-channel EMMI system in laboratory conditions and in clinical settings on human subjects.

The article is organized as follows: Section II analyzes the non-equilibrium sampling technique using a 2-channel equivalent circuit with the goal of understanding the trade-offs introduced by circuit parameters; Section III presents a circuit level and system level design of the wireless, multi-channel EMMI recording system; Section IV-C presents the system characterization results using an 8-channel prototype which is then used to acquire EMMI signals under clinical settings from human subjects, and Section V concludes the article with a brief discussion on future directions.

II. NON-EQUILIBRIUM MULTIPLEXING AND SIGNAL SAMPLING

Non-equilibrium multiplexing is analyzed in this section using a two-channel signal acquisition system whose equivalent circuit is shown in Fig. 3(a). The time-varying biopotentials

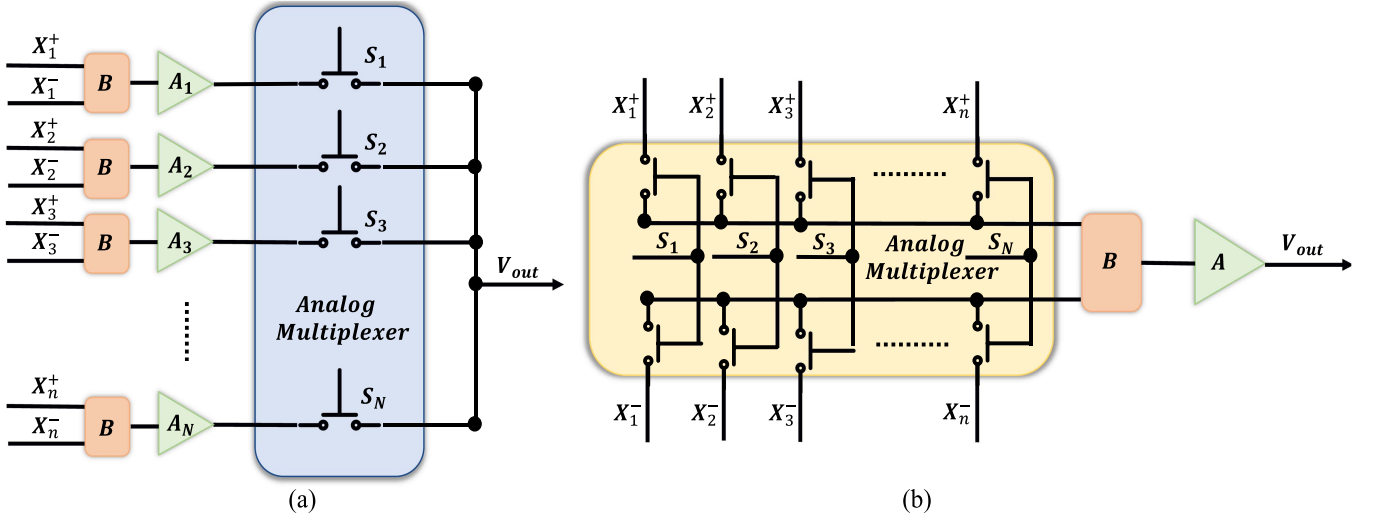


Fig. 2. System architectures used for multi-channel acquisition of biopotentials: (a) using independent amplifiers and filters before analog multiplexing; and (b) using a differential analog multiplexer before a high-bandwidth instrumentation amplifier and a precision filter.

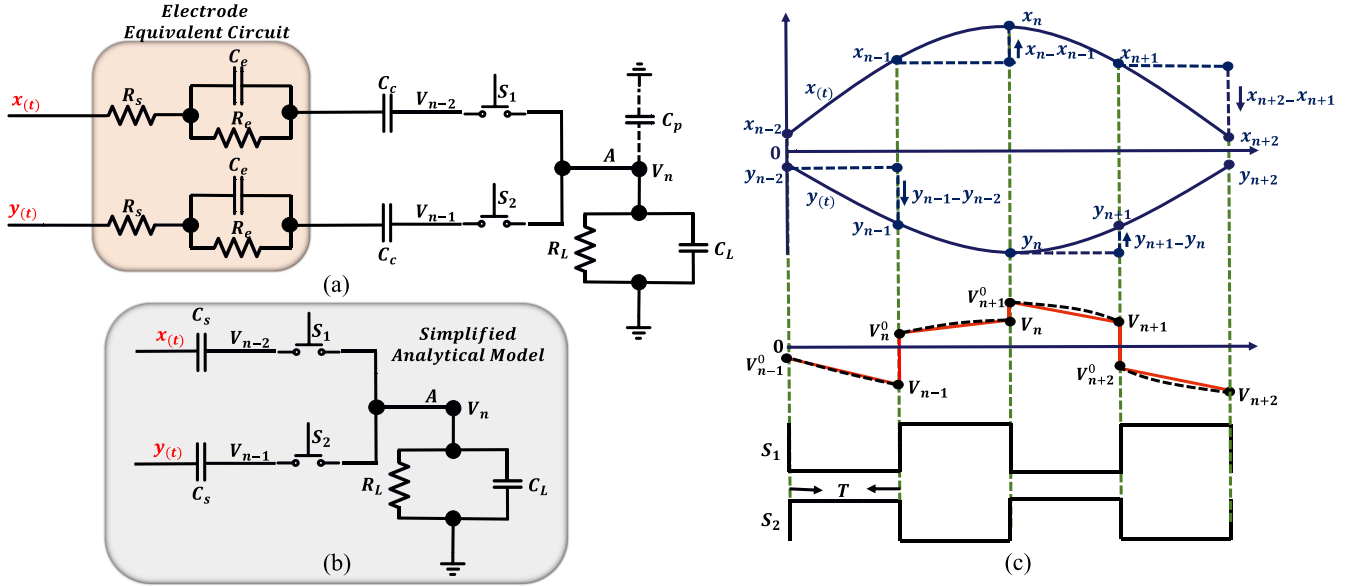


Fig. 3. A simplified two-channel model to understand the effect of non-equilibrium sampling of two input signals $x(t)$ and $y(t)$: (a) Equivalent circuit of the two-channel system comprising of the recording electrodes and the input stage of the amplifier; (b) simplified equivalent circuit that ignores the electrode series resistances; and (c) Timing diagram showing different variables and approximation used for the analysis. Both V_n^0 and V_n represent a linear approximation to the true response which is shown by the dotted line.

$x(t)$ and $y(t)$ are acquired using recording electrodes which in Fig. 3(a) are represented by their simple equivalent circuits [26]. The coupling capacitance C_C is used here for DC decoupling. The biopotentials $x(t)$ and $y(t)$ are assumed to be single-ended and are referred with respect to a common ground, however, the practical implementation described later in Section III-A uses a fully differential topology. The input stage of the instrumentation amplifier is modeled as a simple RC load formed by R_L and C_L in Fig. 3(a). The objective of this analysis is to understand the effect of switching on the voltage sampled at the load capacitance C_L (or at the input of the amplifier) denoted by

A in Fig. 3(a). In literature, the value of R_s is in the range of hundred ohms whereas the value of R_e is in the range of 100k to Mega ohms [27]. Also, since the circuit will be operating in a non-equilibrium (non-steady) state, we can neglect the effect of R_e and R_s compared to the impedance due to C_e and C_c the circuit in Fig. 3(a) can be simplified as Fig. 3(b) where the coupling capacitance C_S is given by

$$C_S = \frac{C_e C_c}{C_e + C_c}. \quad (1)$$

Note that the effect of the parasitic capacitance C_p in Fig. 3(a) has been combined with C_L . We specifically consider the n th sampling instant in Fig. 3(b) when the switch S_1 is closed (or switch S_2 is open) for a fixed duration T . This is illustrated in the timing diagram shown in Fig. 3(c). During the time-interval $(n-1)T \leq t \leq nT$, the voltage at node A V evolves according to the first-order differential equation

$$(C_S + C_L) \frac{dV}{dt} + \frac{V}{R_L} = C_S \frac{dx}{dt} \quad (2)$$

subject to the initial condition which is the voltage V at the beginning of the time-instant nT denoted by V_n^0 . We will also denote the value of V at the end of the nT time-instant as V_n . The initial value V_n^0 can be estimated using the magnitude of the biopotentials sampled at time instants nT which we denote as $x_n, y_n, n = 1, 2, \dots$ and is shown in Fig. 3(c). If the time-constant $R_L C_L$ is assumed to be significantly larger than the time to switch between channels, then the effect of charge leakage through R_L can be ignored. Applying charge conservation we can estimate V_n^0 as

$$C_s(V_n^0 - x_{n-1}) + C_L V_n^0 = C_s(V_{n-2} - x_{n-2}) + C_L V_{n-1}, \quad (3)$$

Note that the charge conservation (3) takes into account two consecutive switching cycles as shown in Fig. 3(c), since the input x_n is sampled after every two clock cycles. The RHS of (3) represents the total charge on the capacitor C_s connected to $x_{(t)}$ and the load capacitor C_L at the end of the $(n-2)T$ cycle. The LHS of the (3) denotes the charge on C_s and C_L at the beginning of the nT clock cycle.

V_n^0 represents the initial voltage at the sampling node A at the instant of the beginning of n th sampling period. Then according to the conservation of charge, we have (3) leads to

$$V_n^0 = \frac{C_L V_{n-1} + C_S V_{n-2}}{C_L + C_S} + \frac{C_S}{C_S + C_L} (x_{n-1} - x_{n-2}) \quad (4)$$

To simplify our analysis, we linearize the time evolution of V in (2) within a switching cycle as $dV/dt \approx (V_n - V_n^0)/T$ as shown in Fig. 3(c). Similarly, $dx/dt \approx (x_n - x_{n-1})/T$ which when applied to (2) leads to

$$\frac{1}{\omega_{lp} T} [V_n - V_n^0] + V_n^0 = \frac{1}{\omega_{bp} T} [x_n - x_{n-1}] \quad (5)$$

where ω_{lp} is given by

$$\omega_{lp} = \frac{1}{R_L(C_S + C_L)}, \quad (6)$$

and ω_{bp} is given by

$$\omega_{bp} = \frac{1}{R_L C_S}. \quad (7)$$

Considering that the biopotential $y(t)$ is sampled at alternate time-instants $n-1, n-3, \dots$, the discrete-time recursion for the two-channel system can now be completely described by

$$V_n + [\omega_{lp} T - 1] V_n^0 = \frac{C_S}{C_S + C_L} [x_n - x_{n-1}] \quad (8a)$$

$$V_{n-1} + [\omega_{lp} T - 1] V_{n-1}^0 = \frac{C_S}{C_S + C_L} [y_{n-1} - y_{n-2}] \quad (8b)$$

where

$$V_n^0 = \frac{C_L V_{n-1} + C_S V_{n-2}}{C_L + C_S} + \frac{C_S}{C_S + C_L} (x_{n-1} - x_{n-2}) \quad (9a)$$

$$V_{n-1}^0 = \frac{C_L V_{n-2} + C_S V_{n-3}}{C_L + C_S} + \frac{C_S}{C_S + C_L} (y_{n-2} - y_{n-3}). \quad (9b)$$

Equations (8a) and (8b) can be used to understand the trade-offs involved in non-equilibrium switching. If $\omega_{lp} = 1/T$ in (8), then the effect of V_n^0 and V_{n-1}^0 cancels and hence there is no cross-talk between the measurements V_n and V_{n-1} . This is the ideal condition for non-equilibrium sampling and specifies a choice of C_S and C_L for a given sampling period T . Also, the RHS of (9a) and (9b) requires that $C_S \gg C_L$ to achieve a maximum signal gain. However, from (1), the maximum value of C_S is determined by the electrode capacitance C_e , implying C_L needs to be reduced. This implies that the effect of parasitic capacitance C_p needs to be reduced. Later in Section III-A we show that this can be achieved using active shielding techniques. The size of the capacitance C_S and hence C_c also determines the error introduced by channel charge injection by the switches S_1 and S_2 and generally a large value of C_c needs to be chosen to alleviate the switching artifacts. The other trade-offs involved in choosing the value of C_c include the effect of thermal noise when sampling the input. Due to the presence of multiple trade-offs, it is practically impossible to ensure optimal non-equilibrium sampling condition $\omega_{lp} = 1/T$, hence the approach would require cross-talk compensation and equalization due to V_n^0 and V_{n-1}^0 in (9). We first present the formulation for the two-channel case which is then generalized to an arbitrary number of channels.

If we denote the measured biopotentials $\tilde{x}_n = V_n$ and $\tilde{y}_n = V_{n-1}$ and the discrete-time equation can be written using z-transforms in a matrix form as

$$\begin{bmatrix} \tilde{x}(z) \\ \tilde{y}(z) \end{bmatrix} = \begin{bmatrix} \tilde{V}(z) \\ z^{-1} \tilde{V}(z) \end{bmatrix} = \begin{bmatrix} H_{11}(z) & H_{12}(z) \\ H_{21}(z) & H_{22}(z) \end{bmatrix} \cdot \begin{bmatrix} x(z) \\ y(z) \end{bmatrix} \quad (10)$$

where $x(z), y(z) \in \mathcal{C}$ denotes the z-transform of the sampled biopotentials and $H_{ij}(z) \in \mathcal{C}$ denotes the cross-channel transfer function from j th channel to the i th channel.

We can generalize the two-channel formulation in (10) to an N -channel formulation as

$$\tilde{\mathbf{x}}(z) = \mathbf{H}(z) \mathbf{x}(z) \quad (11)$$

where $\tilde{\mathbf{x}}(z) \in \mathcal{C}^N$ is the N -channel measurement vector, $\mathbf{x}(z) \in \mathcal{C}^N$ are the N biopotentials being acquired and $\mathbf{H}(z) \in \mathcal{C}^N \times \mathcal{C}^N$ represents the cross-channel transfer function matrix. Therefore, using (11), the biopotential signals can be reconstructed according to

$$\mathbf{x}(z) \approx \mathbf{H}^{-1}(z) \tilde{\mathbf{x}}(z). \quad (12)$$

In the measurement Section IV-B we will show how the cross-channel matrix $\mathbf{H}$ can be estimated under the assumption that non-equilibrium switching maintains the linearity of the system.

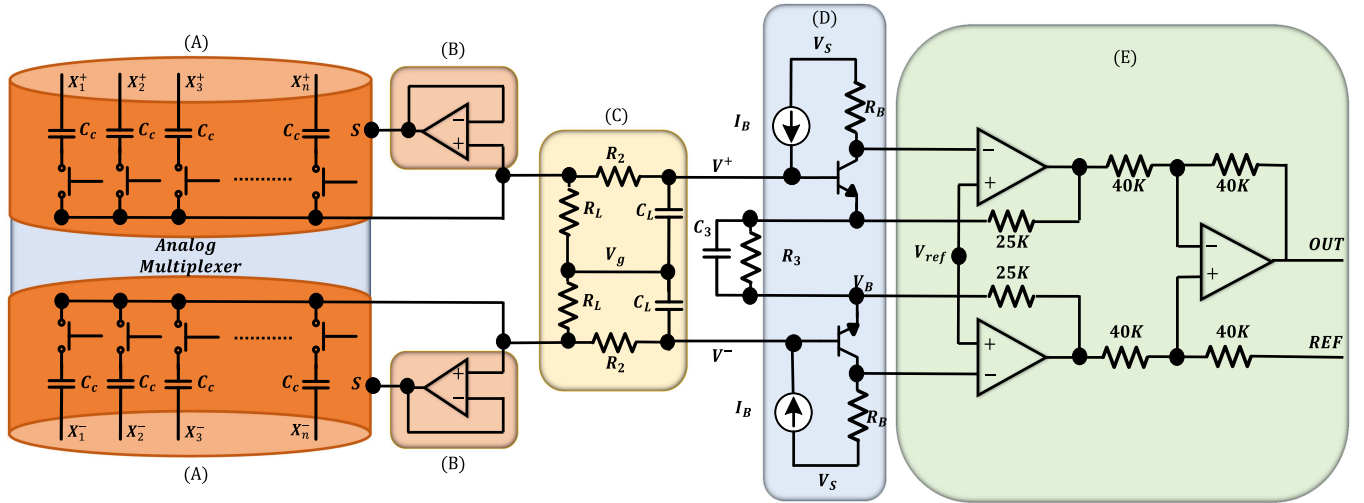


Fig. 4. Circuit implementation of the multiplexed analog frontend comprising of: (a) differential analog multiplexer; (b) active shield circuit; (c) filtering network; (d) input common-mode circuit; and (e) instrumentation amplifier.

III. MULTI-CHANNEL SYSTEM ARCHITECTURE

A. Multiplexed Frontend Circuit

Even though the analysis in Section II is based on a single-ended topology, differential topology is generally considered to be superior from a practical point of view. Differential switching and sampling can potentially alleviate artifacts due to common-mode interference and channel charge injection. Also, the analysis presented in Section II prescribed coupling capacitance C_c whose value should be larger than the equivalent electrode capacitances C_e . For wearable and flexible electrodes, the capacitance values could be in the range of 100 nF- 1 μ F [22], [23], [24], [25], implying that C_c is difficult to implement using integrated circuits-based solution. Fig. 4 shows a differential analog multiplexer circuit A that is connected to a differential passive filter C. A virtual ground is formed at node V_g with respect to which the differential filter output $V^+ - V^-$ is generated. Note that since the value of C_L is relatively large $\approx 1 \mu$ F, the value of R_L could be chosen to be in the range of 1 Mohms to ensure a DC cut-off frequency less than 0.1 Hz. Contrast this to an integrated circuits implementation [14] where pseudo-resistors with large resistances 1 G Ω are required to ensure a low DC cut-off. The active shield circuits B in Fig. 4 formed by the unity-gain buffers ensure that the voltage of the shield S encapsulating the electrodes, coupling capacitances, and switches track the voltage at the input of the filter. This ensures that the effect of any parasitic capacitances C_p along the signal path (shown in the equivalent circuit in Fig. 3) is canceled out. The value of R_2 was chosen such that the higher cut-off frequency is set to 16 kHz. An instrumentation amplifier is used for measuring $V^+ - V^-$, and in Fig. 4(e) we show the equivalent circuit corresponding to the Texas Instrument(TI) INA819 [28] amplifier. In the circuit 4(D), the tail current of a super-beta bipolar input stage of INA819 has been used to set the input common-mode voltage and to ensure a very high input impedance. Since INA819 is a commercial product, we have

omitted the operating details here and the readers are referred to [28] for more information. The INA819 also has a large gain-bandwidth product (GBP) and a fast settling time which ensures that a large number of input channels can potentially be multiplexed. For the proposed design, the value of R_3 was chosen to achieve a gain of 40 dB at a bandwidth of 270 kHz.

B. System and Firmware Design

Fig. 5(a) shows the system architecture of an 8-channel prototype designed to acquire EMMI signals. The differential analog multiplexer has been implemented using ADG725 [29] from Analog Devices and the active shield buffers have been implemented using OPA130 [30]. The output of the INA819 instrumentation amplifier is digitized using a 16-bit SAR analog-to-digital converter(ADC), Texas Instrument ADS8689 [31] with a sampling rate (for all 8 channels) of 3.2 kHz. The digitized output from the ADC is processed by a Texas Instrument CC1310 [32] microcontroller. After every 10 switching cycles, the microcontroller generates a data packet comprising 160 samples which is then wirelessly transmitted to a remote data processor acting as a receiver. A 915 MHz wireless interface based on Texas Instrument's proprietary protocol has been used for wireless transmission. As shown in Fig. 5(a) the received wireless data is then demultiplexed into individual channels, filtered using a median filter after which a channel cross-talk compensation algorithm is applied. Details of the compensation algorithm is provided later in Section IV.

The assembled and packaged 8-channel EMMI system is shown in Fig. 5. The system comprises two printed circuit boards (PCBs) which are shown in Fig. 5(b): a motherboard that integrates the amplifiers, ADC, the microcontroller, and the radio-frequency (RF) modules; and a daughterboard that integrates the analog multiplexer and the active shield circuits. This modular design will allow for future expansion of the system to more than 8-channels without modifying the motherboard.

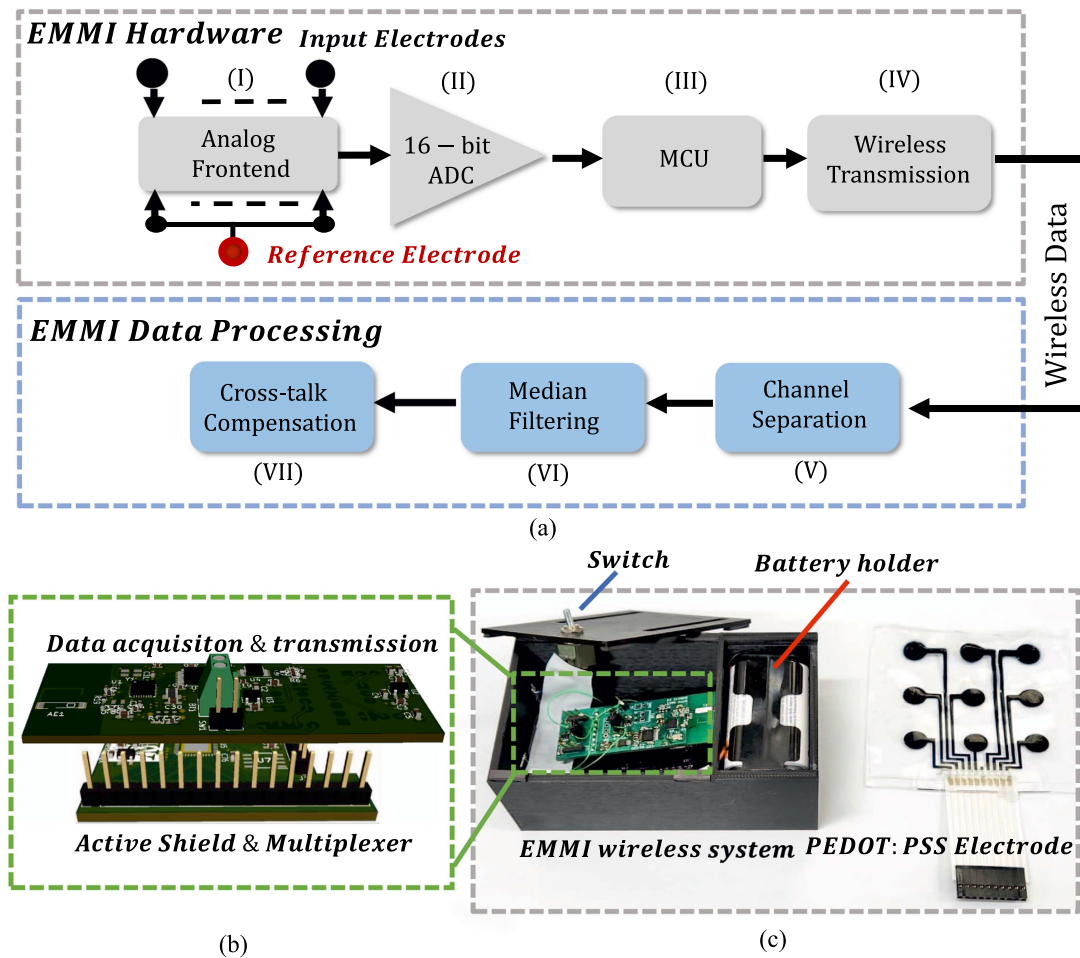


Fig. 5. (a) System level architecture of an 8-channel EMMI recording system comprising of (I) the analog frontend; (II) a 16-bit SAR ADC; (III) a microcontroller; (IV) wireless transmission circuits; (V) software demultiplexer; (VI) a software median filter; and (VII) a software channel cross-talk compensation module. (b) Stacking of the EMMI instrumentation boards where active shielding and analog multiplexing have been implemented on the daughter board and the amplification, the data acquisition, and wireless transmission modules have been implemented on the mother board. (c) Assembled 8-channel prototype which shows the encasing with the instrumentation boards, batteries, and the interface to the flexible PEDOT:PSS electrodes.

The instrumentation interfaces with a flexible electrode array shown in Fig. 5(c). The electrodes have been fabricated using a polymer blend reported in our prior work [26] comprising polyethylene oxide (PEO) and conductive polymer poly(3,4-ethylenedioxythiophene) polystyrene sulfonate (PEDOT:PSS) on a polydimethylsiloxane (PDMS) substrate, which is flexible enough to be worn over the abdomen. The PEDOT:PSS electrodes can be easily fabricated using a low-cost ink-jet printing process and can achieve high conductivity and large-capacitance per unit area - which ensures robust acquisition of biopotentials. Fabrication of the PEDOT:PSS electrode array and characterization of its electrical impedance has been described in detail in [26] and has been omitted here for the sake of brevity. For the sake of easy manufacture, the electrode array is designed as a 3x3 layout. However, only 8 electrodes are activated to accommodate our 8-channel prototyped system. The prototype shown in Fig. 5(c) also integrates batteries and the system can be turned ON/OFF using a flip-switch. When the system is first turned ON, a visual indicator on the receiver notifies the user if the wireless packets are being received after which the data is

parsed and logged on the receiver for post-processing. Details for the transmitter and receiver firmware have also been omitted here for the sake of brevity.

IV. MEASURED RESULTS

A. Single Channel Characterization

The first set of experiments was designed to verify the linearity of the proposed multiplexed system and to characterize the transmission channel so that the compensation procedure described in Section IV-B can be applied. Using a programmable waveform generator we applied a 1 Hz sinusoidal signal to only one of the 8 channels, where all the other channels were shorted to ground. The power spectral density (PSD) of the received data was then used to characterize the quality of signal acquisition and the transmission channel. The measured PSD is shown in Fig. 6, which clearly shows a peak at 1 Hz. However, the PSD also shows peaks at harmonics of 20 Hz which corresponds to the wireless packet transmission rate. This implies that the wireless signal is either getting coupled to the input of the

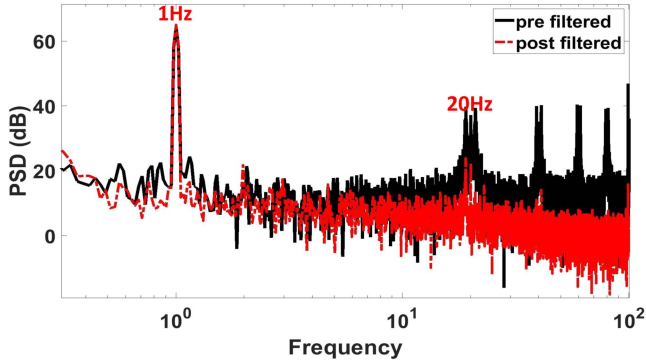


Fig. 6. Power spectrum obtained for the single-tone experiment before and after median filtering.

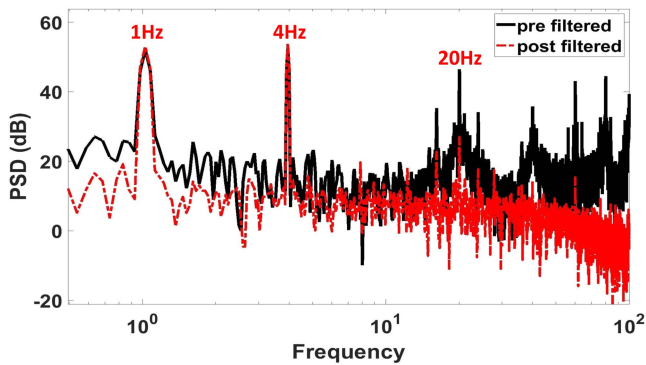


Fig. 7. Power spectrum obtained for the two-tone experiment before and after median filtering.

amplifier due to imperfect shielding or getting coupled due to the current switching during the wireless transmission. Note that this artifact can be potentially alleviated by careful board design, but cannot be completely eliminated. Here we resort to a simple signal processing technique to filter out the wireless transmission artifacts. Note that the interference lies within the frequency band of interest, and therefore, frequency-domain filtering is not very effective. However, wireless transmissions occur in bursts which appear as large voltage spikes in the received data. Thus, exploiting the temporal nature of the bursts and using a simple median filter (window size of 5 samples), could potentially reduce the transmission artifacts. Fig. 6 shows the PSD after median filtering has been applied and clearly highlights the effectiveness of the approach. Applying a median filter also reduces the effect of noise-folding and the PSD plots in Fig. 6 clearly show that the signal-to-noise ratio (SNR) approximately increases from 30 dB to 40 dB. A two-tone experiment was used to determine the linearity of the proposed acquisition system. Two sinusoidal signals at two different frequencies (1 Hz and 4 Hz) were applied to channel 1. The measured PSD in Fig. 7 clearly shows the two peaks at 1 Hz and 4 Hz with no significant harmonics at 3 Hz or 5 Hz. After the RF interference artifacts were compensated using the median filtering technique, the improvement was found to be more profound and demonstrated that the acquisition process can be modeled as a linear system.

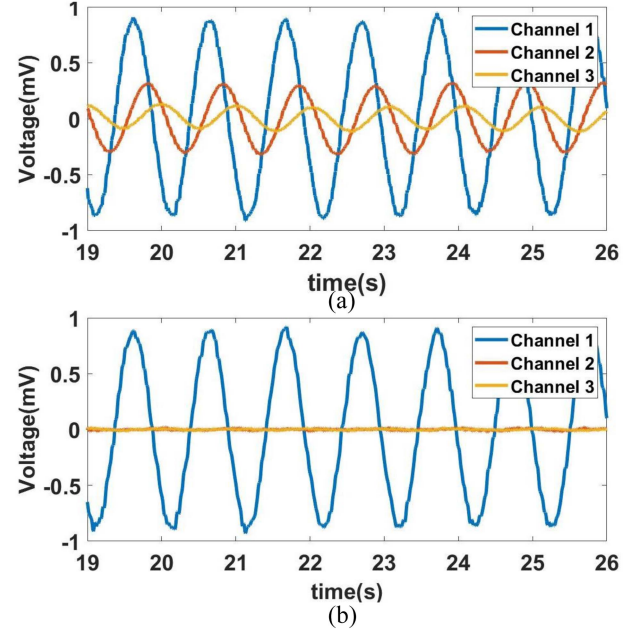


Fig. 8. Single tone experiment used for cross-talk compensation: (a) Recorded output at channels 1,2 and 3, when a 1 Hz sinusoidal signal is applied to channel 1; (b) Reconstructed signals after compensation algorithm is applied.

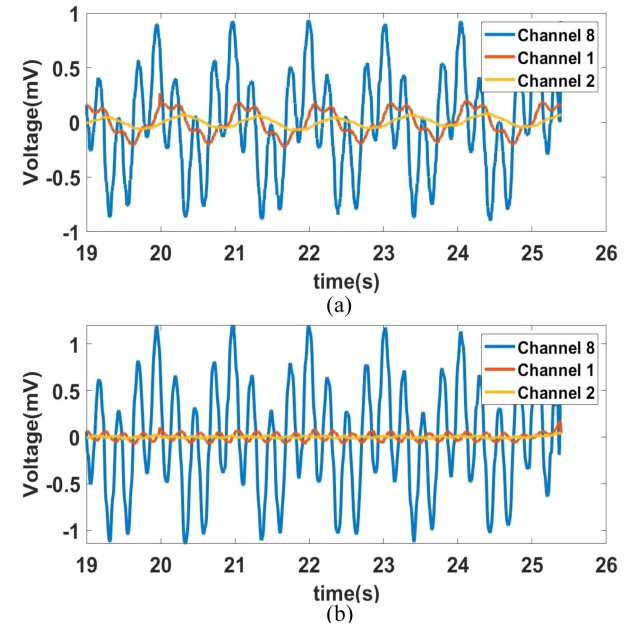


Fig. 9. Two tone experiment used for cross-talk compensation: (a) Recorded output at channels 8,1 and 2, when 1 Hz and 4 Hz sinusoidal signals are applied to channel 1; (b) Reconstructed signals after compensation algorithm is applied.

B. Channel Cross-Talk Compensation

We use the single and two-tone experiments to characterize the cross-talk between the channels. As an example Fig. 8(a) shows the measured output corresponding to channel 1,2 and 3 when a 1 Hz signal is applied to channel 1. The measured result shows the cross-talk but also highlights three important

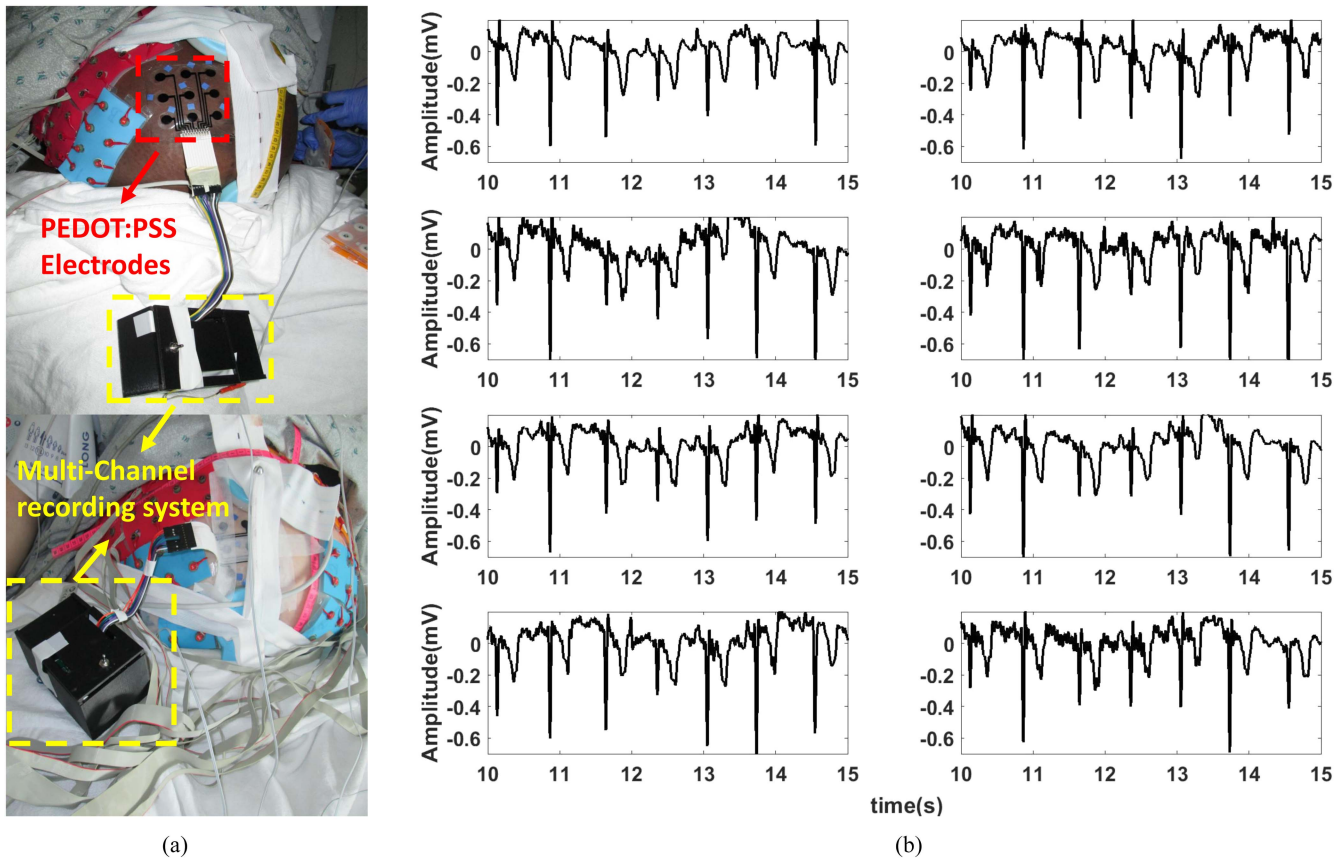


Fig. 10. Human subject study using the prototype 8-channel EMMI system: (a) Experimental setup (showing two subjects) in a clinical setting showing the flexible electrode array interfacing the portable instrumentation; (b) A sample 8-channel recording showing the maternal ECGs.

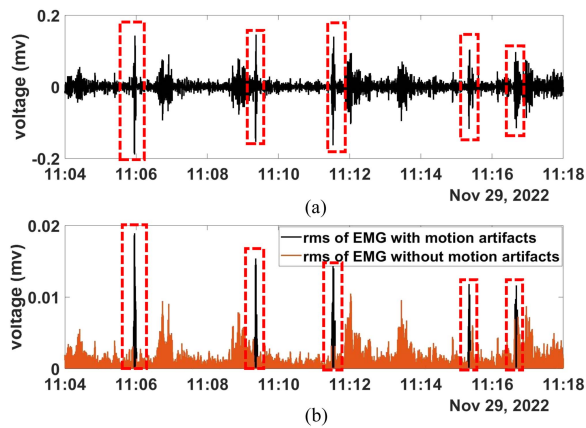


Fig. 11. Extraction of the EMMI signal in the presence of motion artifacts: (a) EMG signal from a single channel after filtering the signal recorded by the 8-channel prototype - segments with significant motion artifacts highlighted in red; (b) RMS of the signal used for identifying motion artifact segments and their removal.

properties that can be used to characterize the cross-channel transfer function matrix in (11):

- The effect of channel cross-talk decreases as the sampling interval between the channels increases. For instance, the system switches to channel 2 after measuring

TABLE I
MEASURED SYSTEM SPECIFICATION

Parameter	Value
Number of Channels	8
Avg. Current Consumption	140 mA
Battery Capacity	1900 mAh
Battery Life Time	13.57 hours
Sampling Rate/Channel	200 Hz
Radio Band	915 MHz ISM
TX Distance	50 m
TX Power	-10 dBm

channel 1, hence the effect of cross-talk from channel 1 to channel 2 will be larger than from channel 1 to channel 3.

- Both the magnitude and the phase of the cross-talk signal are a function of frequency.
- The cross-channel transfer matrix implies that the effect of channel 2 on channel 3 is significantly higher than the effect of channel 2 on channel 1. This artifact is due to the sequence of sampling that is applied for measurement.

Based on these observations, we will consider the cross-channel transfer function matrix of the form

$$H(\omega) = \begin{bmatrix} H_{11} & 0 & \dots & 0 & 0 & H_{1,7} & H_{1,8} \\ H_{21} & H_{22} & 0 & \dots & 0 & 0 & H_{2,8} \\ H_{31} & H_{32} & H_{33} & 0 & \dots & 0 & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & H_{8,6} & H_{8,7} & H_{8,8} \end{bmatrix}. \quad (13)$$

where we have assumed that the cross-talk in each channel arises only from the previous two channels. Each of the terms $H_{ij}(\omega) \in \mathbb{C}$ in the matrix can be represented in its polar form as

$$H_{ij}(\omega) = A_{ij}(\omega)e^{j\omega\delta t}. \quad (14)$$

where $A_{ij} \in \mathbb{R}^+$ is the magnitude and δt is a constant that is a function of the sampling period. To estimate $A_{ij}(\omega)$ and δt we input an ideal sinusoidal signal in one channel while keeping the rest of the channels at zero potential. We further assume that both the magnitude and the phase components have a linear relationship with respect to the input signal frequency. Two different frequencies 1 Hz and 5 Hz were chosen and A_{ij} in (14) was estimated by dividing the peak-to-peak amplitude of channel 2 and 3 output by that of channel 1. Similarly, the time-delay between the peaks was used to estimate the phase-shift in (14). The procedure was repeated to estimate each of the coefficients $H_{ij}(\omega)$ and two different frequencies and an interpolation coefficient was estimated to determine the rate of change of A_{ij} and phase with respect to frequency.

For reconstruction, we computed the discrete Fourier transform of all the output channels. We then separated them into different frequency bins and for each frequency, we applied an inversion procedure using the estimated $H_{ij}(\omega)$. All the reconstructed output spectra for each channel were added and then an inverse Fourier transform was applied to generate the time-domain signal. The number of frequency bins and the size of the reconstruction matrix were chosen based on the apriori knowledge of the signal frequency band (which is 0.1 Hz - 10 Hz) for this application. Fig. 8(b) shows an example of reconstruction for one channel at an input frequency of 1 Hz. The result clearly shows that the cross-channel artifacts have been compensated.

Fig. 9 shows the reconstructed signal for the two-tone experiments where the input signal is a superposition of 1 Hz and 4 Hz sinusoids. Again, the results show that the reconstruction procedure can alleviate the effects of channel cross-talk, however, the compensation is not perfect. We attribute this imperfect compensation to the linear approximation of the magnitude and phase with respect to frequency.

C. Human Subject Experiments

The performance of the prototype 8-channel EMMI system was verified in a clinical environment on patients who were recruited under the human study approved by the Washington University in St. Louis Institutional Review Board (IRB ID No.201612140). The study was conducted on low-risk term labor patients with scheduled induction at the Washington University in St. Louis Medical Center. The wearable system was

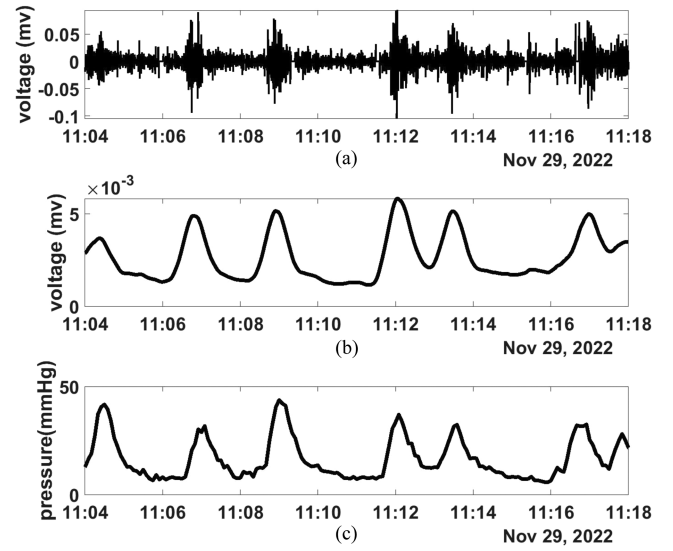


Fig. 12. Extraction of EMMI signals: (a) EMG signal after discarding time-segments with significant motion artifacts; (b) RMS plot of the residual EMG signal; and (c) comparison with the benchmark pressure measurement waveform obtained using a TOCO.

applied after 37 weeks of gestation and was used to record maternal ECG and the EMG signals corresponding to uterine contractions. Fig. 10(a) shows the experimental setup for two of the human subjects where the flexible electrode patch was applied to the abdomen area and the 8-channel multiplexed system was used for simultaneous recording and wireless transmission. For all the experiments, the reference electrode was attached to the patient's shoulder. The wireless receiver and data logger (a laptop) was located in the same examination room. The reconstruction algorithm was applied to the recorded data to extract the multi-channel signal. Fig. 10(b) the 8-channel recordings where the maternal ECG signals can be clearly seen. Note that the magnitude of the ECG signal is ten times larger than the EMG signal corresponding to uterine contractions.

To extract the EMG signals, we applied a filtering procedure described in IV and Fig. 11(a) shows a 14-minute trace of the post-filtered signal. The signal shows EMGs from different sources including motion artifacts which could be significantly stronger than the EMGs due to uterine contractions. However, in real-world clinical settings, these artifacts are unavoidable due to inadvertent patient movements and hence need to be compensated for. Fortunately, the magnitude and temporal nature of the motion artifacts differ from that of the uterine contractions, and hence can be used for discrimination. We start by estimating the root mean square (RMS) of the windowed signals as shown in Fig. 11(b). The motion artifacts (highlighted region) have a higher magnitude than the uterine contraction signals, which enables us to recognize and remove the signal segments. The residual EMG signal is shown in Fig. 12(a) where the uterine contractions are clearly visible as shown in Fig. 12(b). To verify that these signals indeed correspond to the uterine contractions, a tocodynamometer (TOCO) device was used to simultaneously measure changes in pressure (corresponding to the contractions). The TOCO measurements are shown in Fig. 12(c) and clearly correlate to the magnitude of the EMG signal recorded by the

TABLE II
COMPARISON WITH OTHER MULTI-CHANNEL BIOPOTENTIAL ACQUISITION SYSTEMS

Parameter	This work	Intan RHD2132 [34]	BioSemi ActiveTwo [35]	Aziz [36]	Shulyzki [37]	Bagheri [38]
Number of Channels	8	32	280	256	256	1024
Gain	40 dB	45.7 dB	NA	68 dB	53 dB	72 dB
Bandwidth per Channel	34 kHz	10 kHz	0.4 kHz	1 Hz to 5 kHz	10 Hz to 5 kHz	1 Hz to 5 kHz
AFE Power per Channel	240 μ W	271.4 μ W	14.3 mW	4.2 μ W	12.9 μ W	12.9 μ W
AFE Power per Channel (256 channels @ 1 kHz)	7.5 μ W	27.6 μ W	35.8 mW	0.8 μ W	2.6 μ W	2.6 μ W

8-channel EMMI system, thus verifying the operation of the 8-channel system.

V. DISCUSSION AND CONCLUSIONS

In this article we proposed the design of a portable, wireless, multi-channel wearable system for monitoring uterine contractions. The system can therefore be used for monitoring the progression of pre-term labor and can be used in low-resource settings. In Table I we summarize the system specifications of the 8-channel prototype built using commercial-off-the-shelf (COTS) chipsets and whose functionality has been verified in a clinical setting on human subjects. The key novelty of the proposed system is a non-equilibrium multiplexing and signal acquisition technique that should allow the system to be scaled beyond the 8-channel prototype. The compensation technique presented in this article demonstrates that artifacts due to channel cross-talk that arise due to non-equilibrium signal acquisition can be alleviated. Thus, it should be theoretically possible to increase the number of channels according to the mathematical model described by (8)–(9). This would require reducing the sampling period (increasing the sampling frequency) and appropriately choosing the circuit parameters C_C . However, reducing the sampling period would also reduce the signal-to-noise ratio (SNR) since the kT/C_C noise will limit the system, and its magnitude could be comparable to that of the EMMI signal magnitude. Understanding this trade-off for a 128–256 channel system will be a key component of future work. In Table II we compare the system performance of this system with other multi-channel biopotential acquisition systems that have been reported in the literature. Note that most solutions including INTAN RHD2132 use a custom integrated circuit solution, whereas our COTS solution is similar to the popular Biosemi ActiveTwo solution. While the projected results for a 256-channel prototype (last row of the table) show that our COTS-based solution will not perform as well as the prototypes reported in [35], [36], [37], it outperforms both the commercial RHD2132 and the BioSemi solution. Note that a COTS-based solution offers several advantages compared to research-based solutions, namely temperature compensation and superior noise figure. Also, for a large number of channels battery-life is an important factor and in this regard the proposed multiplexing approach has a clear advantage because the power dissipated due to analog multiplexing does not increase significantly compared to the power dissipated when amplifying each channel independently.

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